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Convergence of a Finite Volume Scheme for a Parabolic System Applied to Image Processing

Communication Info

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Abstract

We analyze a finite volume scheme for a nonlinear reaction-diffusion, which is a modified Perona-Malik nonlinear image selective smoothing equation. We establish the existence and uniqueness of solutions to the finite volume scheme, and we prove that it converges to a weak solution. The convergence proof is based on L_2 a-priori estimates and using Kolmogorov's compactness theorem. The numerical simulations are done to verify the efficiency and effectiveness of the scheme.

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